

9. Write a program to implement queue using linked list.

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#include <stdio.h>
#include <stdlib.h>

// Node structure
struct Node {
    int data;
    struct Node* next;
};

struct Node *front = NULL, *rear = NULL;

// Enqueue operation
void enqueue(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

    if(newNode == NULL) {
        printf("Queue Overflow\n");
        return;
    }

    newNode->data = value;
    newNode->next = NULL;

    if(rear == NULL) { // If queue is empty
        front = rear = newNode;
    } else {
        rear->next = newNode;
        rear = newNode;
    }

    printf("%d inserted into queue\n", value);
}

// Dequeue operation
void dequeue() {
    if(front == NULL) {
        printf("Queue Underflow\n");
        return;
    }

    struct Node* temp = front;
    printf("%d deleted from queue\n", front->data);

    front = front->next;
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    if(front == NULL) { // If queue becomes empty
        rear = NULL;
    }

    free(temp);
}

// Display operation
void display() {
    struct Node* temp = front;

    if(front == NULL) {
        printf("Queue is Empty\n");
        return;
    }

    printf("Queue elements are:\n");
    while(temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

// Main function (User Input - Menu Driven)
int main() {
    int choice, value;

    while(1) {
        printf("\n--- QUEUE USING LINKED LIST ---\n");
        printf("1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &value);
                enqueue(value);
                break;

            case 2:
                dequeue();
                break;

            case 3:

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        display();
        break;

    case 4:
        exit(0);

    default:
        printf("Invalid Choice\n");
    }
}

return 0;
}
```